

Template paper for the Robotics: Science and Systems Conference

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Abstract—The abstract goes here.

I. INTRODUCTION

This documents provides a detail explanation of the state-space algorithm that is implemented in this project. The algorithm is developed based on existing state-space algorithm [6][7] and is very similar to the approach used by PLECS [1] and DSpace [3].

II. RECOGNIZE CIRCUIT TOPOLOGY

Although the paper published by PLECS [1] points out how to retrieve state-space matrix from a network topology matrix, but the authors did not give any detail on a systematic method for obtaining those state-space matrix. In Dspace's paper[3], the authors point to a highly influential paper[4] on how to retrieve state-space matrix based on circuit network Topology. In Massarini and Reggiani's paper, the authors points to the book co-author by L.O.Chua and P.M.Lin[2] for methods on retriving state-space matrix base on circuit network Topology. After a detail comparison, the author found many of the ideas by Kuh and Rohrer [4] exist in the book [2]. Thus, the author decide to further develop the algorithm base on the work by Massarini and Reggiani[6], [7]

A. Network Topology

In Chapter 3 of [2], it introduced the idea of "Incidence Matrix", which described circuit topology as a directed graph.

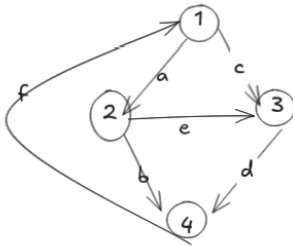


Fig. 1. Network Topology

For example, given the network topology in 1, we have the following incidence matrix in equation 1.

$$\begin{pmatrix} a & b & c & d & e & f \\ 1 & 0 & 1 & 0 & 0 & -1 \\ -1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 1 & -1 & 0 \\ 0 & -1 & 0 & -1 & 0 & 1 \end{pmatrix} \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} \times \begin{pmatrix} ia \\ ib \\ ic \\ id \\ ie \\ if \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad (1)$$

The Node 1,2,3,4 in fig 1 could be viewed as different nodes in the circuit with edge a,b,c,d,e,f represents different electrical components, with the direction represents conventional current flow(from positive to negative). In other sense, the directed graph reflects the KCL of the circuit.

Then, the reduced Incidence matrix A is obtained from Incidence Matrix by "deleting any one row"(typically removing the Ground(GND) Node) to form a reduced incidence matrix, as shown in 2.

$$\begin{pmatrix} a & b & c & d & e & f \\ 1 & 0 & 1 & 0 & 0 & -1 \\ -1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 1 & -1 & 0 \end{pmatrix} \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} \times \begin{pmatrix} ia \\ ib \\ ic \\ id \\ ie \\ if \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad (2)$$

Given A contains the complete information of KCL equations, we can further partitioned the matrix into $A = [A_T \ A_L]$ where coumns of A_T (Tree) correspond to the tree branches of a chosen T , and columns of A_L (CoTree) correspond to the links A_T is an invertable matrix. To correctly reorder A into A_T and A_L , one can simply find the pivot columns of A and form a pivot columns of A as A_T and other non-pivot columns as A_L matrix.

For example, 2 is reorder into 3, where columnner d, e, f are the pivots of 2

$$\begin{pmatrix} d & e & f & | & a & b & c \\ 0 & 0 & -1 & | & 1 & 0 & 1 \\ 0 & 1 & 0 & | & -1 & 1 & 0 \\ 1 & -1 & 0 & | & 0 & 0 & -1 \end{pmatrix} \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} \times \begin{pmatrix} id \\ ie \\ if \\ ia \\ ib \\ ic \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad (3)$$

Although matrix A_a described the KCL law which claims that the sum of currents enter a node equals to sum of current

goes out of a loop, the autho of [2] pointed out a more general form of the KCL states that the algebraic sum of all currents through a cutsets, from one part to the other, is zero at all times. In Chapter 3.4, the author gives detail examples of how to find the D (cutset matrix). But for the purpose of this document, all we need is equation 4 from page 151 of [2].

$$D = (A_T)^{-1}A \quad (4)$$

In Chapter 6.5.2 of [2], the author further expand on the idea of cutset equations(KCL) as shown in 5, where a is the number of port in Tree, z is number of non-port element in Tree, y is number of port in Cotree and b is number of nonport element in Cotree.

The detail of port vs nonport element can be found in Chapter 6.2 of [2], but for our purpose, we just need to remember that Capacitors, Inductors, Voltage/Current source are being classified as port element, and Resistors and all dependent source are classified as non-port element.

For example, a=2, z=1, y=1, b=2 could happen on a D cutset matrix where the first two column are voltage, source and a capacitor, then follow by a resistor, then follow by a inductor with 2 dependents voltage columns.

$$\begin{pmatrix} a & b & c & d \\ I_{aa} & 0_{az} & D_{ay} & D_{ab} \\ 0_{za} & I_{zz} & D_{zy} & D_{zb} \end{pmatrix} \times \begin{pmatrix} i_a \\ i_z \\ i_y \\ i_b \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (5)$$

At this point, this is where the paper by Massarini and Reggiani[7] start develop based on the (6) which obtained from (6-66) and (6-67) of [2], where the las row gives the nonport elements charafcteristics.

$$\begin{bmatrix} Day_{11} & 0 & 1 & 0 & D_{ab} & 0 \\ 0 & -(D_{zb})^T & 0 & 1 & 0 & -(D_{ab})^T \\ \hat{F}_{iy} & \hat{F}_{vz} & \hat{F}_{ia} & \hat{F}_{vb} & \hat{F}_{ib} & \hat{F}_{va} \end{bmatrix} \times \begin{bmatrix} i_y \\ V_z \\ i_a \\ v_b \\ i_b \\ v_a \end{bmatrix} = 0 \quad (6)$$

III. EXTENSION OF THE WORK

At the end of section 2in [7], the author mentioned both M^0 and Q is not full rank when there exist dependency between state variables. However, the author did not give any specific details in handling those situation while maintaining the number of state variables consistent through out the simulation.

In equation 1 and 6 of [7], the author gives the Linear Time Invariant LTI equations to be 7

$$\begin{aligned} M_f \hat{x} &= Ax + Bu \\ y &= C_1 \hat{x} + Cx + Du \end{aligned} \quad (7)$$

$$\hat{x} = M_f^{-1}(A \times x + B \times u), M_f = IE \quad (8)$$

where E is the Capacitance/Inductance of the correspond element and I is the identity matrix of size equals to number

of inductor and capacitor. If M is full rank, it implies the result of

$$IE = M_f$$

is invertible, which simplifies the state-space equation in (8) to

Consider the following state-space form for an hypothetically RLC circuit. Clearly, M_f is always invertible regardless the value of L and C.

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} L & 0 \\ 0 & C \end{bmatrix} \hat{x} = \begin{bmatrix} -0.001 & 10 \\ 1 & 2 \end{bmatrix} x + \begin{bmatrix} 2 \\ 3 \end{bmatrix} u$$

However, the equation in (8) no longer holds when M is no longer full rank. In this case, several possible solution is possible.

In [10], it recognized such system as a "Singular Perturbed Systems." In [10], the authors proposed filling M matrix with small ε value . However, the process of determine this ε is often a iterative, trial-error process.

On the other hand, in [9], [8], and [5] views such singular system as descriptor state space system. In both [9] and [8], it proposed to use either shuffle algorithm, Weierstrass-Kronecker Decomposition or Drazin Inverse method to transform the descriptor state space system into standard state-space form. However, the transformation involves redefine the state variable as

$$\begin{bmatrix} z \\ w \end{bmatrix} = V \times x_i$$

where V is a matrix and x_i is the independent state in the system. In other words, the new states is linear combination of independent states of the original system. While this method works well for large system containing parallel capacitors or series inductor, it is not the perfect solution for state-space simulation. Because for every switch/diode change, it transformed into a new topology with own set of M, A, B matrix, which also changes the definition of state variable. Given the new topology's initial state value depends on the old topology state value, it would be an issue if the old topololgy's z defined by different x_i from the new topology's z.

In this project, I end using a similar simplification that is similiar to shuffle algorithm. In (9), x_1 represents the independent state variable and x_2 represents the dependent state variable.

$$\begin{bmatrix} M_{f11} & M_{f12} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} B1 \\ B2 \end{bmatrix} u \quad (9)$$

Given A22 is invertable, which is the case when there does not exist CE-loop/I-J cutset [2] in the circuit,

$$x_2 = -A_{22}^{-1}A_{21}x_1 - A_{22}^{-1}B_2u \quad (10)$$

and

$$M_{f11}\hat{x}_1 + M_{f12}\hat{x}_2 = A_{11}x_1 + A_{12}x_2 + B_1u \quad (11)$$

which simplifies to

$$[M_{f11} - M_{f12}A_{22}^{-1}A_{21}]\hat{x}_1 = (A_{11} - A_{12}A_{22}^{-1}A_{21})x_1 + (B_1 - A_{12}A_{22}^{-1}B_2)u \quad (12)$$

//TODO: add explain why is full rank and invertible
Then, (12) simplifies to

$$\hat{x}_1 = [M_{f11} - M_{f12}A_{22}^{-1}A_{21}]^{-1}[(A_{11} - A_{12}A_{22}^{-1}A_{21})x_1 + (B_1 - A_{12}A_{22}^{-1}B_2)u] \quad (13)$$

In addition, by taking derivative on both side of (10), it turn to be

$$\dot{\hat{x}}_2 = -A_{22}^{-1}A_{21}\hat{x}_1 - A_{22}^{-1}B_2\hat{u} \quad (14)$$

Given the simulation has a relative small iteration step, $\hat{u}$ is effectively almost 0, which simplify the (14) to

$$\dot{\hat{x}}_2 = -A_{22}^{-1}A_{21}\hat{x}_1 \quad (15)$$

Furthermore, from [7], the author define the output matrix of the system as

$$Qy = C_1\hat{x} + Cx + Du \quad (16)$$

which simplify to

$$Qy = C_1E^{-1}[Ax + Bu] + Cx + Du \quad (17)$$

From [7], dependency in output will cause Q not being a full rank matrix. However, we can still simplify the equation to standard form by finding the minimum norm of the Q matrix

IV. STATE SPACE ITERATION

As mentioned in [3], we can discretized the continous state-space representation using Pade approximation into an discreted state-space representation. By doing pade approximation, we remove the need to calculate exponent of matrix and turn the iteration process into Matrix Vector operations only.

V. CONCLUSION

The conclusion goes here.

ACKNOWLEDGMENTS

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